Architectural Patterns for Integrating LLMs into User-Facing Applications

Lessons from a language-learning platform

MIRCEA LUNGU, IT University of Copenhagen, Denmark

CESARE PAUTASSO, University of Lugano, Switzerland

An LLM placed behind a feature in an existing application has to meet expectations it was not designed for: an answer in the time a user will wait, at a cost the feature can carry, in a form the surrounding code can consume. How to reconcile a slow, costly and non-deterministic model with those expectations is little documented. Products built around the model, such as chatbots, agents and retrieval-augmented pipelines, have taken most of the attention. Fitting a model into an application that already exists is a different problem. We present a catalogue of nine recurring architectural patterns for that case, grounded in a year and a half of LLM integration work on Zeeguu, an open-source language-learning platform with several hundred monthly active users. The patterns are grouped into three themes, using the LLM efficiently, trusting its output, and managing change over time, and described in the standard context, forces, solution, and consequences format. Each pattern names the Zeeguu features it came from and what corroborates it elsewhere, whether published work, a tool that ships the same mechanism, or another team's engineering account, and says where we found no such use outside Zeeguu.

Additional Key Words and Phrases: large language models, software architecture, design patterns, LLM integration, cost, latency, quality assurance

1 Introduction

While there is growing literature on building products whose central function *is* the model — chatbots, agents, and retrieval-augmented generation (RAG) pipelines, which practitioners often call *LLM-native* — and on using LLMs for code generation, there is surprisingly little guidance on the **software engineering challenges of integrating LLMs as components into existing interactive applications** where real users expect fast, reliable, and trustworthy responses.

We expect this integration to become the common case: over time, more and more existing user-facing applications will adopt LLMs not as standalone chatbots but as components working behind the scenes to improve the user experience. This is why we present a set of patterns that highlight the challenges and opportunities such integrations raise.

Indeed, LLMs have a unique combination of properties:

- **expensive**: billed per token, with a large fixed prompt re-paid on every call;
- **slow**: responses take seconds, not the milliseconds an interactive interface expects;
- **non-deterministic**: the same input can return a different answer;
- **rapidly evolving**: models are released and retired on the vendor's schedule, every few months;
- **imprecise**: they make mistakes, from a malformed response to a confident but wrong answer; and
- **general-purpose**: they can attempt almost any task expressed in text.

Putting a component with these properties into a live, user-facing system pulls a design in two directions at once: the model is slow, costly and non-deterministic, while users and the surrounding code need answers that are fast,

Authors' Contact Information: Mircea Lungu, IT University of Copenhagen, Copenhagen, Denmark, mircea.lungu@gmail.com; Cesare Pautasso, University of Lugano, Lugano, Switzerland.

affordable and well-formed. Competing pressures of this kind are what the pattern literature calls **forces**, and every pattern in this paper is an account of which forces it resolves and at what cost.

Over the past year, we have been integrating LLMs into Zeeguu[1], an open-source platform for personalized language learning that helps users learn foreign languages by enabling them to read authentic online content (real articles, not textbook exercises). Through this work, we have identified a set of architectural patterns for LLM integration, several of them corroborated by documented use in other systems. Because these pressures come from the LLM itself and from the ordinary demands of a live application, not from anything specific to language learning, we expect them, and the patterns that resolve them, to recur wherever LLMs are integrated into user-facing systems.

The remainder of the paper is organised as follows. Section 2 describes how the patterns were identified and how the resulting claims were checked. Section 3 introduces Zeeguu, the platform that grounds every example. The catalogue then follows in three themes, using the LLM efficiently (Section 4), trusting its output (Section 5), and managing change over time (Section 6), each pattern given in the same format: context, example, problem, forces, solution, consequences, known uses, notes. Section 7 asks what makes these patterns specific to LLMs, and the paper closes with related work, limitations, and conclusions.

2 Methodology

The patterns come from one production system. This section states how they were found, so that a reader can judge what the evidence supports.

The corpus. Zeeguu (Section 3) has been in continuous operation since 2017 and currently serves several hundred monthly active users across eleven languages. Between January 2025 and August 2026 we added LLM-backed functionality across roughly a dozen of its features. The patterns below draw on ten: article simplification, CEFR difficulty assessment and summarisation, grammatical correction of simplified text, topic classification, multi-word expression detection, on-demand translation escalation, audio-lesson scripting, audio-lesson topic validation, exercise sentence generation, and exercise sentence validation. Those features, and the code and commit history behind them, are the raw material.

Where the patterns came from. We did not set out to mine patterns. The design decisions came first, made under ordinary delivery pressure as each feature was built. Only after a year and a half of this did we ask which of those decisions had been reusable: which solutions we had reached for more than once, and which we would reach for again in a different system. The patterns presented here are our answer to that question, and the conditions below are what we then required of each.

What earned a place here. Two conditions. Inside Zeeguu, a solution had to appear in at least two features that reached it independently, rather than one copying the other; those instances are named with each pattern, and the source is public, so a reader can check them. Outside Zeeguu, we looked for the same solution in another system, a practitioner report, or published work, and each pattern states the closest evidence we found and how close it is. Applying these conditions changed what this paper contains: solutions we had expected to present did not survive them.

The evidence base for known uses. Corroborating uses come disproportionately from grey literature: engineering blog posts, provider documentation, and open-source code, rather than peer-reviewed papers. This is deliberate, and at this point unavoidable. The practice is roughly three years old and largely industrial, and the peer-reviewed record has

[1] https://zeeguu.org

not caught up (Section 8). Garousi et al.'s criteria for admitting grey literature as evidence in software engineering are met here on several counts: the topic is not covered by the formal literature, practitioners are the population that holds the knowledge, and the field is moving faster than the publication cycle. We mark each known use with its source type, so a reader can weigh it.

What this is not. This is not a systematic literature review, and not a controlled study. It is one team's reading of one system in one domain. The claim we make is that the pressures a pattern resolves come from the LLM's own cost, latency and unreliability, and from what a live application owes its users, neither of which is specific to language learning. Whether the patterns hold elsewhere is an empirical question, and Section 9 returns to it.

3 Case Study: Zeeguu

Zeeguu is an open-source language learning platform built around the idea that learners benefit most from engaging with comprehensible [12] and authentic [8] content in their target language. Rather than relying on artificial textbook texts and exercises, Zeeguu helps users find real articles (news, blog posts, and other web content) tailored to both 1) their *level* and 2) their *interests*. Then, based on the words they don't understand, it generates personalized vocabulary exercises and audio lessons.

The platform recommends articles in the learner's target language based on their proficiency and reading preferences, making it **easy to find material that is both engaging and appropriately challenging**. If a text is personally compelling but too difficult, Zeeguu simplifies it to the learner's level using LLMs.

When users encounter unfamiliar words or phrases while reading, they can get contextual translations on the fly from several translation providers, so the reading experience remains fluid and uninterrupted. One alternative, available on demand, comes from a state-of-the-art LLM offering a more contextually nuanced option (the "Ask an AI" escalation described below).

Every translation a user requests is logged by the system, which over time builds a **detailed model of the learner's vocabulary knowledge**, tracking which words they know, which ones they struggle with, and how well they've retained previously encountered vocabulary.

Based on this evolving learner model, Zeeguu **generates interactive vocabulary exercises and audio lessons** that focus on the words that matter most for each individual learner, rather than following a generic curriculum. The exercises use the context in which the word was originally encountered, based on the assumption that if the original text was compelling to the learner, examples drawn from it will be too.

In essence, Zeeguu unifies reading, translation, learner modeling, and practice into a coherent pipeline, with the learner's own reading interests as the primary driver.

Zeeguu also supports teacher accounts: a teacher can assign texts to a class and follow each student's reading and exercises. Because a teacher's judgement is more authoritative than an individual learner's, their corrections act as a higher-trust signal in the learner model.

3.1 The Pieces

A handful of Zeeguu-specific concepts recur across the patterns. They are collected here once, so a pattern can point back to a single definition rather than re-explaining them.

3.1.1 CEFR Levels. The Common European Framework of Reference grades language proficiency on an ordered six-level scale, from A1 (beginner) to C2 (mastery). Zeeguu uses it in two directions: to estimate how hard an article is, and to simplify an article down to the level of a given learner.

3.1.2 Article Simplification. When an article is compelling but too hard, Zeeguu rewrites it with an LLM to easier CEFR levels, producing one simplified version per level below the original. It runs both on demand, when a reader opens an article that has not been simplified yet, and ahead of time, over the crawled feed for some articles deemed likely to appeal to readers.

3.1.3 Crawling. Zeeguu builds its article recommendations by crawling news sites and blogs multiple times per day; this steady feed of freshly crawled articles is where most ahead-of-time LLM work happens (CEFR assessment, simplification), off any reader's critical path. On demand, readers push their own content in: a browser extension sends any article to Zeeguu for study, and on mobile the system share sheet sends any web page to be made interactive and studied within the application.

3.1.4 Multi-Word Expressions. A multi-word expression (MWE) is a group of words whose meaning is not the sum of its parts, for example *break the ice*. Zeeguu detects them so a learner can translate the phrase as a unit rather than word by word. Detection runs on a dependency parse (Stanza [17], an open-source NLP parser). An LLM is called for one narrower case: translating an expression whose parts are separated in the sentence.

3.1.5 Meanings and The Learner Model. A *meaning* is a word paired with one particular translation of it. A meaning is linked to the original context in which it was seen, but can also be associated with other example sentences in which the word carries that same translation. Every translation a learner requests is logged as a meaning, building a model of which meanings they know and which they struggle with, since one word can have several meanings. Meanings are classified (by frequency, CEFR level, and phrase type: single word, collocation, idiom, or expression) and drive which exercises and lessons the learner sees. Vocabulary training trains *meanings*, not words.

3.1.6 Translation. While reading, a learner gets an instant contextual translation that is generated with the help of multiple parallel translation APIs. If they all agree, the translation is inserted above the word. If they disagree, a popup surfaces the disagreement along with the option to escalate to an LLM through the "Ask an AI" action. This is one extra, more expensive step, offered for the cases where it is unclear which of the translations is correct.

3.1.7 Audio Lessons. Zeeguu generates personalized audio lessons: an LLM writes a short script built around the words a learner is studying, and text-to-speech synthesizes the audio. Both steps are slow and costly, so lessons are pre-computed for recently active learners. Once a learner chooses a topic (European history) or a situation (talking to an elderly neighbour on the stairway), a new lesson is generated each day, conditional on the learner having listened to the previous day's lesson.

3.1.8 Vocabulary Exercises. Exercises are built from the words a learner has looked up. They use example sentences drawn from the learner's own reading where possible, and LLM-generated, then LLM-validated, sentences otherwise. Original context is used only where possible, because not all contexts in which a learner encountered a word suit an exercise: some are too long, others do not make sense outside their original context.

3.2 Reach

Zeeguu currently serves over 300 monthly active users across 11 languages (Danish, Dutch, English, French, German, Greek, Italian, Portuguese, Romanian, Spanish, and Swedish), with peaks exceeding 400 during the academic year[2]. It is publicly available to try: the web app is at zeeguu.org[3] with the invite code `zeeguu-beta`, and the mobile apps can be downloaded from the iOS and Android app stores.

4 Using the LLM Efficiently

An LLM call is slow and metered, so a recurring question in any integration is how *not* to make the call, or how to make each one count. The patterns in this section keep the model's cost and latency off the user's path: paying a large fixed prompt once across a batch rather than once per item, reaching for the LLM only when a cheaper tool falls short, letting a classical stage gate it so it runs only where its judgment is needed, and computing likely-needed results ahead of time so the model never runs while a user waits. The unifying force is that the LLM is the expensive, slow component and the cheapest call is the one that is never made.

[2]Monthly active users are defined as users with any learning activity (exercises, reading, browsing, audio lessons, or translations) in a given month. Live statistics are available at: https://api.zeeguu.org/stats/monthly_active_users
[3]https://zeeguu.org

4.1 Prompt Amortization EFFICIENCY PATTERN

Batch many items into one call, so the fixed prompt is paid once.

4.1.1 Context. Many LLM calls share the same shape: a large, fixed instructional prompt (rules, output format, examples) wrapped around a tiny variable input, and re-sent in full on every call.

The work is offline and nobody is blocked on any single result, so an item can wait for others to join it.

4.1.2 Problem. Many calls send a prompt far bigger than the item it carries, and send it again for every item, so most of what is paid for is the same instructions, repeated. How can a system keep its detailed prompt without paying for it once per item?

4.1.3 Problem in Zeeguu. Zeeguu makes twenty-one kinds of LLM call. In most of them, a fixed template of a few hundred to several thousand characters wraps an input many times smaller (Figure 1). The translation-validation prompt[4], for example, wraps 5,096 characters of rules around a word, its translation and the sentence it came from. The templates also keep growing as rules are added: all four in the figure are longer than their first version (marked by the vertical tick).

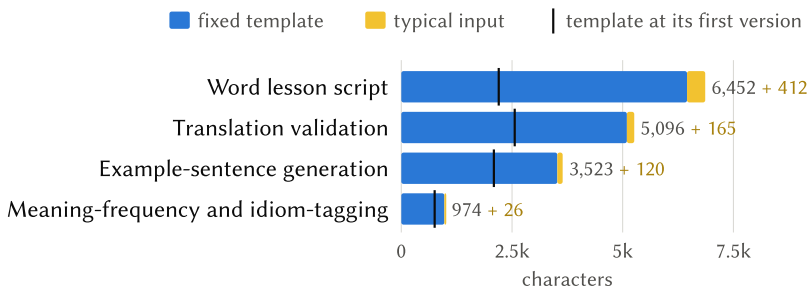


Fig. 1. Fixed prompt template (blue) and typical input (yellow), in characters, for four of Zeeguu's LLM calls at commit c853d76 (October 2026). The black tick marks each template's size at its first version.

4.1.4 Forces.

- **Detailed prompts work better.** The more precisely the rules describe the task, the more accurate and consistent the output. *(pulls toward big prompts)*
- **Cost follows the tokens sent.** Providers charge per token, and locally hosted LLMs use energy in proportion to the tokens they process. Every call sends its whole prompt again, so a big prompt is paid for once per item. *(pulls toward sending fewer tokens)*
- **Quality drops with more items per call.** The more items one call handles, the worse the model can handle each one: it might skip some, mix them up, or shorten its answers. With many small inputs this starts after about 15 to 20 items, long before the context window is full; with full-length outputs, after only a few. How much depends on the model and the task, and only measuring tells. *(pulls toward small calls)*

[4] https://github.com/zeeguu/api/blob/c853d767/zeeguu/core/llm_services/prompts/translation_validator.py#L8-L94

4.1.5 Solution. Put several items into one call, so the prompt is paid once per call instead of once per item,[5] and stop adding items before the quality of the answers drops. There are two ways to do it:

- ***Fan-in* batching** packs many independent inputs into one prompt, spreading a large instructional preamble across the whole batch.
- ***Fan-out* batching** produces many outputs from a single input, emitting one section per output and collapsing several requests into one.

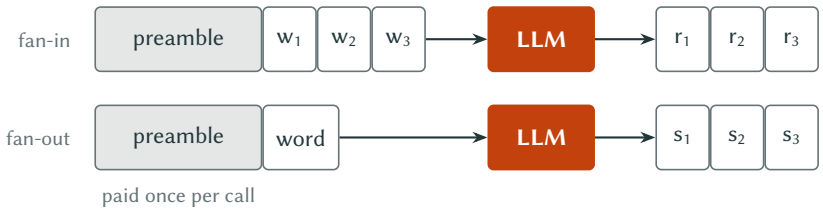


Fig. 2. Fan-in packs many inputs behind one preamble; fan-out asks for many outputs (here, several example sentences for one word) from one. Either way the preamble is paid once per call.

4.1.6 Solution in Zeeguu. Zeeguu uses both forms, on two of the calls in Figure 1:

- **Fan-in: meaning-frequency and idiom-tagging.** Zeeguu stores each word a learner saves together with its translation, a pair it calls a *meaning* (Section 3.1.5). An offline job sends about 15 of them per call and gets back for each one how common it is and whether the phrase is worth studying as a unit: an idiom such as *break the ice* is, a random pair such as *doctor for* is not.[6]
- **Fan-out: example-sentence generation.** One call asks for three example sentences for a word, so its template is paid once for all three.[7]

Fan-out also has a case where Zeeguu went back to one call per item. It used to ask for an article to be simplified (Section 3.1.2) to several levels in a single call: a B2 article came back at B1, A2 and A1 at once. Writing several versions in one answer, the model shortened them and merged paragraphs. An evaluation showed that one call per level simplifies much better, so Zeeguu now pays for the extra calls.[8] For that task, the batch size that works is one.

4.1.7 Consequences.

- **Cost and latency amortize with batch size.** The fixed preamble is paid once per call instead of once per item, so per-item token cost *and* wall-clock latency fall roughly inversely with the batch size.
- **A batch size to tune, and to tune again.** The size that works depends on the model, the prompt and the items, and only measuring finds it. Nothing signals when a new model or a longer prompt has moved it, so it has to be measured again after each change.

[5]Two provider features sound similar but do something else. *Batch APIs* (OpenAI Batch, Anthropic Message Batches) discount large asynchronous jobs, but every request in the job still pays for its full prompt. *Prompt caching* does discount a prompt that repeats across calls, but each item still makes its own call and waits for its own answer. It also works only under conditions: the cached prompt expires if no call reuses it soon enough (five minutes by default on Anthropic), and short prompts are not cached at all. As of October 2026 Anthropic's minimum is 512 tokens, about 2,000 characters, so the meaning-frequency and idiom-tagging template (about 1,000) is too short.

[6]The batched tagging prompt, `create_batch_meaning_frequency_and_type_prompt` in the `zeeguu/api` repository, run by `tools/reclassify_unclassified_phrases.py`.

[7]The example-generation prompt, `format_prompt` in the `zeeguu/api` repository, called with `count=3`.

[8]The old multi-level prompt, `get_adaptive_simplification_prompt`, is now reached only from a manual tool, `tools/simplify_article.py`. The split, its evaluation and its cost are in zeeguu/api#773.

- **Outputs have to be matched back to their items.** A batched answer can skip an item or stop short, so the code checks which items got a usable output, rather than assuming they line up, and sends the rest again (*Defensive Output Parsing*).
- **Fan-in adds a wait.** Items sit until a batch fills, so each result arrives later than it would on its own. This is why the pattern needs work that nobody is waiting on.

4.1.8 Known Uses.

- *The fan-in direction is prior art.* Packing many independent inputs under one shared prompt is the published **batch prompting [3]** technique (Cheng, Kasai & Yu, EMNLP 2023), which cuts token and time cost roughly inverse-linearly with batch size. This pattern's contribution is to add the *fan-out* direction and to frame both as a single move: amortizing one fixed preamble.
- The *fan-out* direction maps directly to structured-output calls that emit several keyed results at once, and to the OpenAI/Anthropic multi-output `n` parameter (which requests several completions per call).

4.1.9 Related Patterns. *Anticipatory Precomputation* (computing likely-needed results ahead of time) makes work offline, and offline work can wait for a batch to fill. *Task Fusion* is the mirror image: it shares one input across many tasks, where this pattern shares one set of instructions across many items.

4.2 Escalate to the LLM EFFICIENCY PATTERN

Serve the common case with a cheap tool; call the LLM only when the user questions the result.

4.2.1 Context. A feature can be produced either by a cheap specialized tool (a translation API, a classical classifier) or by an LLM. The tool's answer is shown to the user. The tool gives no score saying how sure it is, but the user often notices when its answer feels off.

4.2.2 Problem. Most results from the cheap tool are good enough, and paying for the LLM on those is waste. A few are not, and on those the LLM is worth its cost. How can a system call the LLM only for the results that fall short?

4.2.3 Problem in Zeeguu. In Zeeguu, translation APIs (from Google, Azure, and DeepL) translate the words a reader taps. Most of the time the translation is enough. Sometimes the reader is not convinced: the gloss does not fit the sentence, or a compound or idiom seems to mean something else here.

4.2.4 Forces.

- **The LLM does better on the hard cases.** On the inputs where the cheap tool falls short, such as idioms and compounds, an LLM usually does better. *(pulls toward using the LLM)*
- **Every LLM call costs money and time.** An LLM call costs more than the cheap tool and is slower, so making it on every request would make the common case slower and costlier. *(pulls toward using the cheap tool)*

4.2.5 Solution. Show the cheap tool's result first, and give the user a way to ask for a better one. On that signal, call the LLM with the original input and, where it helps, the cheap tool's output and the user's feedback, so the escalation refines rather than restarts. The LLM may answer a different question than the tool did: Zeeguu's translation gives a word, the escalation explains it.

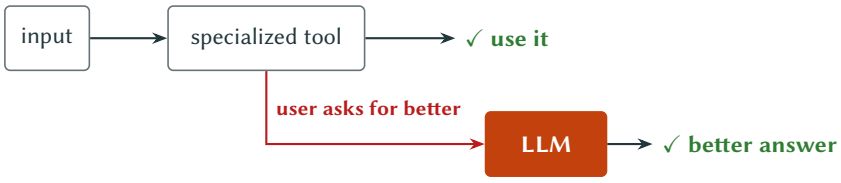


Fig. 3. The cheap tool serves the common case. The LLM runs only when the user asks for a better result.

4.2.6 Solution in Zeeguu. When the reader is not convinced by a translation, they tap *Explain* in the translation menu, and an LLM explains what the word means in this sentence, at the reader's level. The explanation works as a check: the reader can judge the translation instead of having to take it on trust.

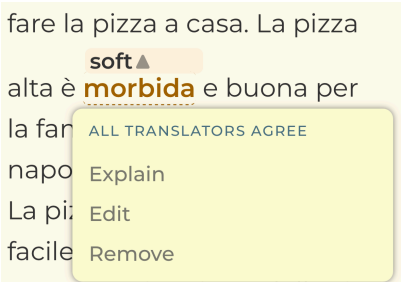


Fig. 4. In Zeeguu, the inline translation is the primary path; when the reader wants more, they escalate to an LLM on demand via the "Explain" option.

Explain does not send the LLM the translation the reader was shown, because the gloss misled it. For the Danish *narrøv* glossed as "foolish ass", the model reasoned backwards from the gloss and split the word into *nar* + *øv* ("donkey") rather than *nar* + *røv*. With the gloss in the prompt, 0 of 3 runs were correct; with only the word and its sentence, 3 of 3.[9] Whether the cheap tool's output helps the escalation or misleads it has to be measured.

4.2.7 *Consequences.*

- **Cheap in the common case.** The LLM runs only when a user asks, so most requests never incur its cost or latency.
- **The user carries the routing.** They must notice the option and use it, and it takes space in the interface. Some readers of earlier versions argued that the user should never have to ask; this pattern accepts that cost because nothing else can tell a poor result from a good one.
- **The better answer comes later.** The LLM starts only when the user asks, so its answer takes a full LLM call to arrive. The user is not left empty-handed meanwhile: they already have the cheap result.

4.2.8 *Known Uses.*

Duolingo Max[10] (2023): after an exercise the learner sees the usual correction, and can tap *Explain My Answer* to get a GPT-4 explanation of why their answer was right or wrong. The explanation is generated only for learners who ask.

4.2.9 *Related Patterns.*

- **Model cascade.** In ML serving, a model cascade has the same shape, cheap tier first and expensive tier on demand, but escalates automatically when the cheap model's own confidence is low. **FrugalGPT [2]** (Chen, Zaharia & Zou, 2023) escalates when a scorer rejects the cheap answer, and **RouteLLM [16]** (Ong et al., 2024) trains a router that sends only the hard queries to the strong model. The difference is who decides: a score, or the user. Where the cheap tool gives a usable score, a cascade can decide without asking, and the two can be combined.
- ***Fail-Fast Provider Chain*** moves to another provider when a call fails. Here the cheap tool did not fail: its answer was not good enough for this reader.
- **Routing.** When the hard cases can be recognized before the cheap tool runs, send them straight to the LLM. Zeeguu does this for multi-word expressions (Section 3.1.4) whose parts are separated in the sentence, as *call*

[9] `explain_selection` and the measurements in `prompts/word_explanation.py`, in the `zeeguu/api` repository.
[10] https://investors.duolingo.com/news-releases/news-release-details/duolingo-max-shows-future-ai-education

and *back* are in *I'll call you back*: a translation API would get a fragment without its context, so the whole sentence goes to the LLM and the API is never called.[11]

[11] `translate_separated_mwe` in the `zeeguu/api` repository.

4.3 Anticipatory Precomputation EFFICIENCY PATTERN

Generate the LLM results users will likely need before they ask.

4.3.1 Context. A user-facing feature needs an LLM result, but the model takes seconds while the user expects the result in real time, and *which* results a given user will need next is predictable from their behaviour.

4.3.2 Example. The vocabulary exercises a learner practices are built from the words they looked up while reading, each paired with a machine translation. At reading time an imperfect translation is low-stakes: the reader is only making sense of the text, and can pick a better option from the alternatives menu. But once the platform selects a word for the learner to *learn*, they will drill that word-translation pair over many days, so its correctness suddenly matters. A regular cron job looks ahead: it finds the words each learner is due to study next and validates them with an LLM, so that when the exercise module asks for new words, the vetted ones are ready straight away. If none are precomputed yet (the learner translated a few words and jumped straight into exercises), the validation runs in real time.

An even costlier instance: audio lessons (Section 3.1.7). Generating a personalized audio lesson is more expensive again: an LLM writes the lesson script (fed the learner's past lessons so a recurring topic, a "talking to a neighbour" lesson they have had before, comes back fresh rather than repeated), then text-to-speech synthesizes the audio, several seconds of work no learner should wait through. A nightly job precomputes the next lessons for recently active learners (prioritized by how recently they practiced), on the assumption that someone who has been studying will be back for the next one. When they return, the lesson is already waiting. Only a learner who switches to a new topic waits, briefly, while that day's first lesson on it is generated on demand.

4.3.3 Problem. How can an LLM-quality result reach the user's critical path without a wait for the model, when upcoming needs are often predictable?

4.3.4 Forces.

- **Latency.** Real-time users expect an answer in about 200ms, but an LLM can take several seconds depending on the prompt and deployment, so it cannot run on the critical path. *(pushes toward precomputing)*
- **Wasted spend on misses.** Precomputing spends tokens on results that may never be requested, so a poor predictor pays for nothing and still misses. *(pushes toward precomputing only high-probability needs)*
- **Precomputed results must be held until they are used.** A precomputed answer has to persist from the moment it is generated to the moment it is asked for, so the working set grows with the number of users, the length of the lookahead, and the size of each result: bytes for a validated translation, megabytes for a generated audio lesson. *(pushes toward a short lookahead and a prioritized subset of users)*
- **Predictability.** The pattern is available only where upcoming needs can be forecast from behaviour; the better the behaviour model, the more of the work can move off the critical path.

4.3.5 Solution. Anticipate likely user needs and precompute LLM results offline (e.g., via cron jobs), so results are available instantly when needed. The system designer should model user behaviour in order to predict their LLM needs. Keep an on-demand path for cold or mispredicted requests, so a miss degrades to a normal wait rather than a failure.

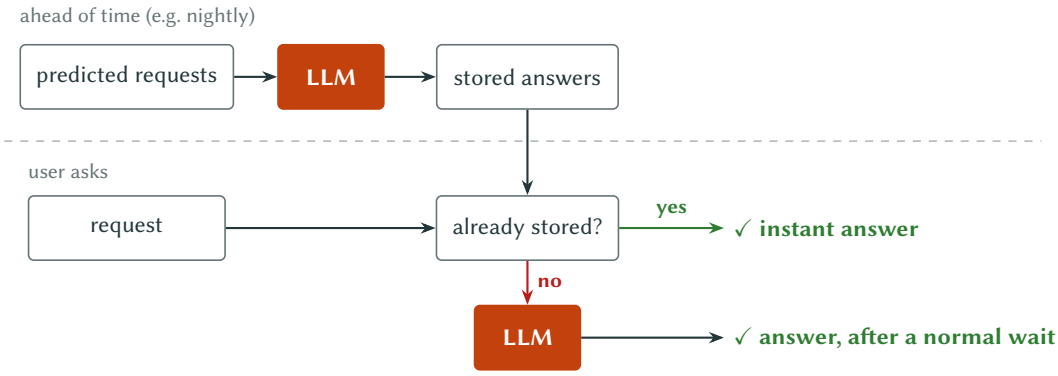


Fig. 5. Ahead of time, the LLM answers the requests the system predicts and the answers are stored. A predicted request is then served from the store with no LLM call; an unpredicted one falls back to an on-demand call, a normal wait rather than a failure.

4.3.6 *Consequences.*

- **Zero latency at request time.** The LLM's cost and multi-second latency are paid entirely off the critical path; when the user acts, the result is already waiting.
- **Precomputing pays for guesses that miss.** It spends tokens on results that may never be requested, so the value depends on the accuracy of the behaviour model: a poor predictor wastes spend *and* still misses.
- **Needs a reliable "what" and "when" signal, plus a fallback.** It applies only where upcoming needs are predictable; cold or mispredicted requests still need an on-demand path. Composes with *Prompt Amortization* (offline results can be batched).

4.3.7 *Known Uses.*

- **Yelp**[12] precomputes LLM query-understanding responses for high-frequency ("head") search queries into a key/value store, "caching (pre-computing) high-end LLM responses for only head queries", reaching 95% of traffic for review-highlight expansions, so the expensive model never runs on the critical path.
- **Instacart**[13]'s Intent Engine serves high-frequency search queries from a precomputed cache of LLM outputs, leaving only ~2% of queries to a real-time model.
- **ColBERT [11]** (Khattab & Zaharia, SIGIR 2020) precomputes contextualized passage embeddings for the whole corpus offline, so query time runs only cheap late-interaction matching: the canonical "precompute the expensive representation ahead of time" (the pattern's shape applied to retrieval rather than generation).

[12] https://engineeringblog.yelp.com/2025/02/search-query-understanding-with-LLMs.html
[13] https://tech.instacart.com/building-the-intent-engine-how-instacart-is-revamping-query-understanding-with-llms-3ac8051ae7ac

5 Trusting LLM Output

An LLM's output is only probabilistically correct, and only probabilistically well-formed, yet it flows into code and data that assume it is neither wrong nor malformed. The patterns in this section guard that boundary: refusing to trust the shape of a response, catching content errors with a second, narrower check, recording how far each stored piece of output has been trusted. The unifying force is non-determinism: the same prompt can return a different, or malformed, answer, so correctness has to be enforced around the model rather than assumed from it.

5.1 Defensive Output Parsing TRUSTING OUTPUT PATTERN

Check the shape of every LLM response in code, and repair it if needed.

5.1.1 Context. An LLM is asked for output in a fixed shape (JSON, a delimited record), and the parsed result feeds code on the critical path. Some providers can enforce a schema and some cannot, and even an enforced schema does not guarantee a response the code can consume.

5.1.2 Example. Multi-word-expression (Section 3.1.4) detection asks the LLM for a JSON array (groups of word positions in a sentence) but refuses to blindly trust that it will get one.

- `_parse_response` first strips any markdown code fence (a triple-backtick code block the model often wraps the JSON in), tries `json.loads`, and on failure regex-extracts the last JSON array in the text (models tend to add a preamble), parses that, and distinguishes a legitimately empty result (`[]`) from a parse failure.
- If nothing parses, it logs the raw text and returns `[]` rather than raising.
- A separate `_validate_groups` step then checks the parsed shape (token indices in range) before anything downstream uses it.

The same layered try / extract / validate / fall-back appears in the translation validator, the simplification (Section 3.1.2) service, and the example generators.

5.1.3 Problem. How can a malformed response, one that code cannot safely consume, be kept from turning into a failed request?

5.1.4 Forces.

- **Format compliance is never fully guaranteed, and a naive parse fails the request.** Without an enforced schema the model can wrap the output in a code fence or add a preamble. With one, the response can still be cut off at the token limit or replaced by a refusal, and the schema language cannot state every constraint the code relies on (an index that must point into the input). Either way the parsed result feeds downstream code, so an unhandled slip turns into a failed request.
- **Provider schema enforcement is free where it exists, but not everywhere.** Constrained decoding removes syntax slips at the source, yet a cheaper, local, or fallback model in the chain may offer only a JSON mode or nothing at all. *(pushes toward relying on the provider for syntax, and keeping code for the rest)*
- **Strictness in the prompt is tempting but brittle.** Formatting failures invite ever-longer prompt instructions, which are hard to test and still only probabilistic. *(pushes toward prompt-stuffing)*
- **Strictness in parsing code is deterministic and testable**, at the cost of code to write and maintain. *(pushes toward defensive parsing)*

5.1.5 Solution. Do not trust the raw output's shape.

Where the provider can enforce a schema (structured output), request it: syntax slips then disappear at the source. Where it cannot, repair the syntax in code: strip known wrappers, attempt a plain parse, and if the model wrapped the output in prose, find the expected structure inside the text and parse that.

In both cases, check in code what the schema cannot guarantee: that the response is complete, that it is not a refusal, and that it satisfies the structural constraints the code relies on (types, ranges, required fields, references into the input).

On failure, degrade gracefully (a default, a skip, a retry, or the next provider in a fallback chain of LLM vendors) instead of raising.

Keep the strictness in code, where it is deterministic and testable, rather than in ever-longer prompt instructions. The question here is only whether code can consume the output; whether consumable output is *correct* is the concern of *Reject and Reprompt* and *LLM-Checking-LLM*.

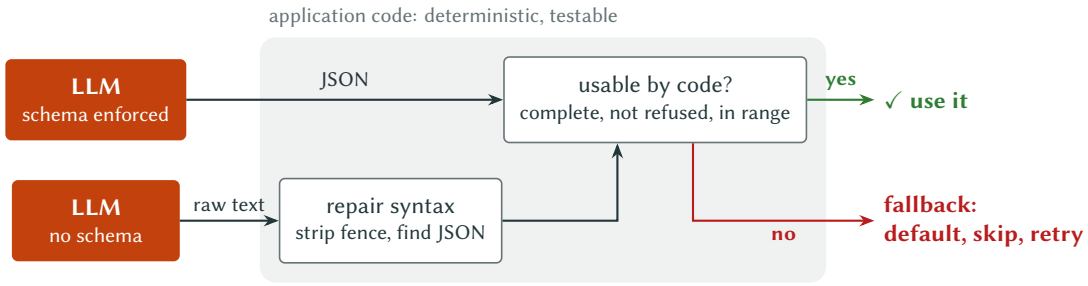


Fig. 6. A provider that enforces a schema sends JSON straight to the check; output from one that does not has its syntax repaired in code first. Either way, code checks what no schema guarantees (the response is complete, not a refusal, and its indices point into the input), and a response that fails degrades to a fallback instead of raising.

5.1.6 *Consequences.*

- A malformed response degrades a single result instead of failing the request: the layered parse recovers what it can, and a clean fallback (a default, a skip, a retry, the next provider) covers the rest.
- The parse layer is code to write and maintain, and it has to be kept in step with the prompt and the provider as the output format drifts.
- Provider structured output removes the syntax repair where it is offered, but not truncation, refusals, or the constraints its schema language cannot state, so the checks shrink rather than disappear.

5.1.7 *Known Uses.*

- **G-Research**[14]'s code-review tool treats LLM output as "unverified input": it normalises responses by "removing wrappers before parsing" (some providers return bare JSON, some wrap it in markdown fences), validates every finding against an authoritative rule index (invalid findings are "rejected outright"), detects truncation via the `length` finish reason and retries with a lower cap, and sends structurally-invalid output back to the model for one repair pass.
- **Honeycomb**[15]'s Query Assistant parses the LLM output, "correct[s] it (if it's correctable)," validates it, and instruments parse vs. validation errors separately before running the query.

5.1.8 *Notes.*

- Composes with a one-shot retry and with a provider fallback chain (a parse failure can trigger the next provider).

[14] https://www.gresearch.com/news/building-a-code-review-tool-the-llm-patterns-that-actually-work/
[15] https://www.honeycomb.io/blog/hard-stuff-nobody-talks-about-llm

- *Partly absorbed by the providers.* When the pattern was first written, every layer was application code. Since 2024 providers enforce schemas by constrained decoding (OpenAI[16] from August 2024, Anthropic[17] from November 2025), which removes the fence-and-preamble repair for those providers. What their documentation still leaves to the caller is the remainder this pattern keeps: both can return output cut off at the token limit or a refusal in place of the schema, Anthropic's schemas cannot state `minimum` or `maximum`, and DeepSeek's JSON Output[18] guarantees valid JSON but not a schema, and "may occasionally return empty content". A pattern shrinking as the platform absorbs part of it is the kind of drift the patterns in Managing Change Over Time address.
- Broader than repairing a specific, known formatting defect: this pattern is the stance of not trusting the structure at all.
- *Prior art: error-tolerant parsing.* Recovering the expected structure and skipping the surrounding text is the LLM-output analogue of island grammars [15] (parse the fragments of interest, skip the rest) and other lenient, error-tolerant parsing long used to pull structure out of irregular input.
- *Enablers (not instances).* Validation/repair is widely productized, Instructor[19] (Pydantic + auto-retry), LangChain[20] `OutputFixingParser`, and provider structured-output[21] modes (which still require handling truncation and refusals), but a library that *provides* validation is the mechanism, not evidence of an in-app stance.

[16] https://developers.openai.com/api/docs/guides/structured-outputs
[17] https://platform.claude.com/docs/en/build-with-claude/structured-outputs
[18] https://api-docs.deepseek.com/guides/json_mode
[19] https://python.useinstructor.com/
[20] https://python.langchain.com/api_reference/langchain/output_parsers/langchain.output_parsers.fix.OutputFixingParser.html
[21] https://developers.openai.com/api/docs/guides/structured-outputs

5.2 LLM-Checking-LLM TRUSTING OUTPUT PATTERN

Check an LLM's output with a second LLM call that verifies one property, a narrower task than generating it.

5.2.1 Context. An LLM generates content that will be used or stored, and its output is sometimes wrong in ways a targeted check could catch. Verifying a specific property (grammaticality, factual match, difficulty level) is a narrower task than the open-ended generation that produced it.

5.2.2 Examples. The vocabulary exercises need example sentences for each word, which an LLM generates at the learner's level, an open-ended task (the sentence must be natural, level-appropriate, and actually use the word). A generated sentence can still be wrong in a specific way: it may use the word in a *different* sense than the one being taught. For the Danish *virker* (translated as *seem*), a generated sentence might use *virker* in its other sense, *work/function*. A second, batched LLM call then asks one narrow question of each sentence, whether it uses the word in the intended meaning (Section 3.1.5), and drops the ones that fail. Verifying that single property is far narrower than writing a good sentence from scratch.

A second pair ran on article simplification (Section 3.1.2), and has since been switched off, which is what makes it worth reporting. Rewriting an article to a lower CEFR level is open-ended; whether the result is grammatical is not. A second call read each simplified title and body and returned corrections, which were applied to the text.[22] Between December 2025 and April 2026 we measured what it caught: roughly 96% of title corrections and 72% of body corrections changed two characters or fewer. The check worked. What it found was not worth what it cost, and it is now disabled by default.

The two pairs differ in what happens on failure, which the pattern deliberately leaves open: an example sentence is expendable, so it is dropped and another word used, while a simplified article is the only version at that level, so the check produced a correction instead. But the retired pair carries the sharper lesson. The asymmetry that makes verification cheaper than generation says nothing about whether the errors are worth catching. That is a separate question, and only measurement answers it.

5.2.3 Problem. How can an unreliable generator's mistakes be caught, when a second generator would be just as unreliable?

5.2.4 Forces.

- **Verification is narrower than generation.** Checking one property (is it grammatical? does it use the intended meaning?) has a small answer space and a clear criterion, so a focused checker is more reliable on that property than the open-ended generation was: the *generation-discrimination gap* measured by Saunders et al. [19]. *(pushes toward adding a check)*
- **The checker is itself an LLM.** It can return its own false verdicts, and it costs a full extra call in tokens and latency. *(pushes toward checking only where the asymmetry is large, or where a classical check exists)*
- **The mistakes must be checkable in isolation.** The pattern helps only for errors a targeted, differently-prompted call can catch, not for failures that need the whole generation redone.

5.2.5 Solution. Use one LLM call to generate a result, then a separate, differently-prompted LLM call to check one specific property of it. This escapes the paradox for two reasons. First, verifying one property (is it grammatical? does it use the intended meaning?) has a small, well-defined answer space and a clear success criterion, so an LLM does

[22] `GrammarCorrectionService` in the `zeeguu/api` repository.

it more reliably than the open-ended generation that produced the output. Second, because the checker is prompted differently and asked a different question, its errors are largely *decorrelated* from the generator's rather than shared, so it does not simply repeat the generator's mistakes.

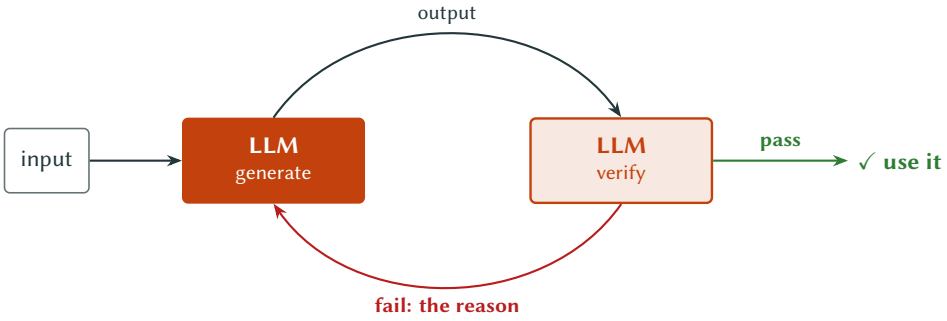


Fig. 7. Verifying is narrower than generating: a second, differently prompted LLM call verifies one property of the first call's output (is it grammatical?), a smaller task it does more reliably. The two calls feed each other: on a failed check the reason goes back to the generator, the loop of Reject and Reprompt; dropping the item or recording the verdict are the other policies.

A failed check still needs a policy, and the pattern does not fix one: drop the result, record the verdict, or *Reject and Reprompt*. Dropping is cheapest and fits where the item is expendable (the vocabulary sentences above are discarded and other words used instead). Recording the verdict alongside the output instead of acting on it is *LLM Content Validation Tracking*. Regenerating with the reason the check gave, within a bounded number of attempts, is *Reject and Reprompt*, whose loop works with any checker, an LLM included.

5.2.6 *Consequences.*

- **A focused check is more reliable than the generation.** Verifying one property is easier than producing the whole output, so the second call catches errors the first introduced, for the price of one extra call.
- **It narrows the error rate, it does not remove it.** The checker is itself an LLM and can return its own false verdicts, and it adds cost and latency.
- **The pair hides behind one interface, not behind a guarantee.** Generation and check can be packaged as a single call, so the rest of the application never sees the judge. That is a convenience, not a contract: the result is still best-effort, exactly as it would be from the generator alone.
- **It pays off only when checking is genuinely narrower than generating.** A check as open-ended as the generation buys little. The verdict composes with *LLM Content Validation Tracking* (record it), and where a classical check exists it is cheaper still.

5.2.7 *Known Uses.*

- *Documented in the literature.* **LLM-as-a-judge [25]** (Zheng et al., NeurIPS 2023) uses a separate strong LLM to score another model's open-ended outputs, now a standard evaluation technique.
- **Self-Refine [14]** (Madaan et al., NeurIPS 2023) has the model critique and refine *its own* output in a separate pass: a same-model variant of the idea, where this pattern instead uses a differently-prompted call.

- **G-Eval [13]** (Liu et al., 2023), shipped in eval frameworks such as **DeepEval**[23], uses a separate chain-of-thought (step-by-step reasoning) judge call to grade generated output.

5.2.8 Notes.

- Where *Defensive Output Parsing* guards that the output is well-*formed*, this guards that a well-formed output is *correct*.
- Distinct from ensemble methods and chain-of-thought: an ensemble averages several generations and chain-of-thought elaborates one, whereas this pattern spends the second call on the cheaper *verification* task instead of more generation.
- *Why the asymmetry holds, and where it stops.* Spending a call on verification rather than generation echoes the intuition behind NP, the class of problems where a proposed answer can be checked quickly even when finding one may be hard: confirming a candidate can be far easier than producing it. The gain is empirical here, not guaranteed, and it depends on the check being a narrow, differently-prompted property. It does not extend to open-ended *intrinsic self-correction*, a model revising its own reasoning with no new signal, which shows no such gain and can even degrade accuracy (as Huang et al. [10] and Stechly et al. [21] find). That is why the check here asks a *different* question through a *separate* call, rather than asking the generator to reconsider.
- *This check may be transient.* It exists because today's models are imprecise enough to need an external verifier. As models improve, or as reliable self-verification moves inside the model itself, the need for a separate checking call may shrink: the pattern answers the current generation's reliability, not a permanent architectural truth.

[23] https://deepeval.com/docs/metrics-llm-evals

5.3 LLM Content Validation Tracking TRUSTING OUTPUT PATTERN

Tag every stored LLM output with its verification state.

5.3.1 Context. LLM-generated content is written into persistent storage alongside human-verified data, and downstream features and users will read it. Some of it has been checked, most has not, and once stored the two look identical.

5.3.2 Examples. A meaning (Section 3.1.5) (a word paired with a translation) can exist at several trust levels: generated by Google Translate (unverified), verified by an *LLM-Checking-LLM* pass (auto-verified), implicitly accepted (a learner drilled it in an exercise without flagging it as wrong), or explicitly corrected by a trusted user, such as a teacher. Each level carries different confidence, and the system can make different decisions depending on the validation state: for example, only including explicitly confirmed translations in exercises, while using auto-verified ones for reading assistance where the stakes are lower.

Simplified articles carry a similar state, arrived at differently. Each stored version has a `broken` flag that keeps it out of what learners are offered, and it is set three ways: an automatic quality check at generation time, an audit tool that finds versions the LLM wrote in the wrong language, and readers themselves, since three reports mark an article broken.[24]

What the state is for differs in each case. A meaning's level is read on every exercise that might use it, and grades how the meaning may be used. A version's flag is read when a reader asks for a level, and excluding a version is also what causes it to be replaced: the on-demand path calls the LLM only when it finds no usable version at that level. One gates use; the other gates use and triggers repair.

5.3.3 Problem. How can trusted and untrusted LLM-generated data be told apart after it lands in the database, so each is used according to its confidence?

5.3.4 Forces.

- **Unmarked data is silently trusted.** Once stored, LLM output is indistinguishable from human-verified data, so without a marker downstream features treat unverified output as ground truth. *(pushes toward tracking trust at all)*
- **Different consumers can tolerate different confidence.** Reading assistance is fine with an unverified translation; a word promoted into drills should be confirmed first. Serving both well means distinguishing levels of trust, not just verified from unverified. *(pushes toward a richer trust spectrum)*
- **Every level is state to maintain.** Each level is another transition to model, set on write and update on every validation event, so too fine a spectrum is maintenance burden and false precision. *(pushes toward the coarsest spectrum that still supports the decisions)*

5.3.5 Solution. Maintain an explicit, queryable validation state for all LLM-generated content in the data model. Never let LLM-generated content silently become trusted data. The validation state may be a simple flag, but more often it is a spectrum or state machine reflecting different levels of trust (e.g., unverified → auto-verified → implicitly accepted → explicitly confirmed by a trusted user; alternatively, one could track the number of users that have validated a given content).

[24] `Article.broken`, `UserArticleBrokenReport`, and `available_simplified_versions` in the `zeeguu/api` repository.

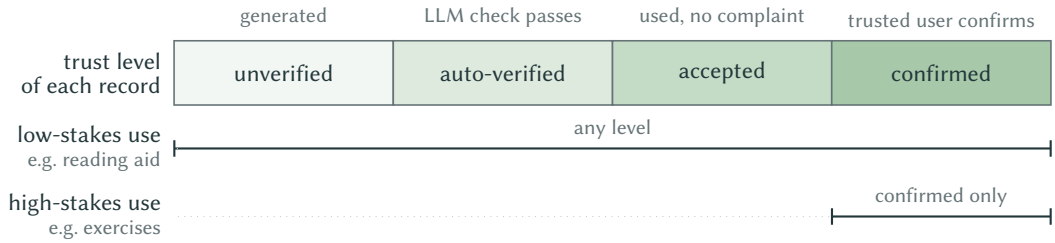


Fig. 8. Every stored LLM-generated record carries a trust level, raised by the events above it. Each use reads only the levels it can tolerate: a low-stakes use accepts any level, a high-stakes one only confirmed records.

5.3.6 Consequences.

- **Each consumer can gate on trust.** Exercises use only confirmed pairs, reading assistance can fall back to auto-verified ones, and unverified output never silently becomes ground truth.
- **The data model carries an extra state to maintain.** It must be set on every write and updated on every validation event, and the right granularity (a flag, a spectrum, or an agreement threshold: N independent users confirming before it counts as trusted) is a domain judgment.
- **It pairs with provenance to describe any stored artifact.** Where *LLM Output Provenance* records how an artifact was made, this records whether it has been confirmed; together they answer both questions about a stored artifact.

5.3.7 Known Uses.

- **Argilla**[25], an open-source platform for annotating and reviewing model-generated data, gives every record an explicit status: a model suggestion starts as *pending*, becomes a *draft* and then *submitted* as someone works on it, and ends up *validated* or *discarded*. A suggestion is never treated as correct until a human has moved it to *validated*.
- **Label Studio**[26], another annotation tool, flags each item as ground truth or not and records whether it has been reviewed, so human-verified "gold" data stays clearly separate from unreviewed model predictions.
- **Snorkel [18]** (Ratner et al., VLDB 2017) attaches probabilistic confidence to programmatically generated labels rather than treating them as ground truth: a confidence spectrum rather than a discrete state machine.

The three tools show the range of options for how finely trust is recorded: a lifecycle with several states (Argilla), a yes/no "gold" flag (Label Studio), and a probability score (Snorkel).

Note. The examples above are data-labeling platforms; a documented, in-product *LLM-output* trust-state lifecycle (as opposed to human-annotation state) remains thinly evidenced: one reason we keep this among the less-settled patterns.

5.3.8 Notes.

- This pattern complements *LLM Output Provenance*. Together they answer two essential questions about any piece of LLM-generated data: how was it produced? and has anyone confirmed it's correct? The right granularity of validation tracking depends on the domain: some systems may need binary (verified/not), others may

[25] https://docs.argilla.io/latest/how_to_guides/annotate/
[26] https://docs.humansignal.com/guide/ground_truths

need multiple validators with agreement thresholds, and others, like Zeeguu, benefit from a trust spectrum that reflects different forms of implicit and explicit user feedback.

- *Implicit signals are weaker than explicit ones.* A word drilled in exercises without ever being flagged is a positive signal, but a weak one, which is why it sits below a correction from a trusted user in the trust order.

5.4 Reject and Reprompt TRUSTING OUTPUT PATTERN

Regenerate a rejected output instead of repairing it, and tell the model what was wrong.

5.4.1 Examples. Audio lesson scripts are generated at a high temperature so the dialogues sound natural, and the same freedom occasionally lets a dialogue drift into the wrong language. A cheap classical language detector checks each generated script. When it rejects one, the script is neither repaired nor silently dropped: it is regenerated with a prompt extended by the rejected output and the constraint it broke. The detector costs almost nothing next to the generation, and the second attempt is conditioned on the specific failure rather than resampling the same distribution.

The same loop runs over article simplification (Section 3.1.2), where a stored summary or a simplified level occasionally comes back in the article's original language instead of the target one. It was extracted into a shared helper once a third call site needed it, and the helper deliberately refuses to decide what happens when the attempts run out, because the three call sites disagree: the assessment-and-summary path drops the summary and keeps the assessment, simplification fails the run or drops the offending CEFR level and keeps the rest, and the provider services return nothing and let their caller decide.[27]

What varies is not the check or the retry but the exhaustion policy, and it varies with how much of the result survives partial failure. Where the output is a bundle of independently droppable pieces, the loop can give up on one and keep the others; where it is a single artifact, giving up means losing all of it.

5.4.2 Forces. A validator can be cheap and classical even when the thing it validates can only be produced by an LLM, so the check costs a fraction of the generation and can afford to run on every output.

Some defects cannot be repaired in place. A stray trailing ellipsis has a deterministic fix; a dialogue in the wrong language does not, and the only route to a correct result is another generation.

A plain retry against a non-deterministic generator carries roughly the same failure probability as the first attempt, so a retry that does not say what went wrong buys another sample rather than a better one.

5.4.3 Solution. Validate generated output with a cheap classical check placed downstream of the LLM. On rejection, regenerate rather than repair, extending the prompt with the rejected output and the property it violated, so the next attempt is conditioned on the failure. Bound the number of attempts, and record a terminal state when they run out. Some failures are permanent for a given input rather than incidental: a model that declines a topic on policy grounds declines it again tomorrow, and a generator that never produces output gives the check nothing to reject. Without a record, a scheduled job rediscovers the same dead end on every run.

[27] `generate_in_language` in the `zeeguu/api` repository.

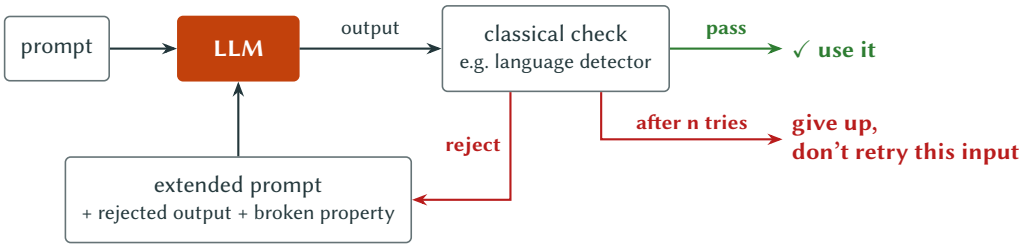


Fig. 9. A cheap classical check sits downstream of the LLM. A rejected output is not repaired: it goes back into an extended prompt, together with the property it broke, so the next attempt is conditioned on the failure. After a bounded number of tries the system records that this input will not get a result, so a scheduled job does not try it again.

5.4.4 *Known Uses.*

- **Zeeguu** validates generated audio lesson (Section 3.1.7) scripts with a classical language detector and regenerates with an extended prompt when the detector rejects a script. The same loop carries the terminal case: when the model refuses a topic outright, judging the example offensive, the loop stops and the word is marked as one no lesson will be generated for, so the nightly job does not retry it indefinitely. *Added in 2026, and too recent to report how often the second attempt succeeds.*
- **Instructor**[28] feeds the Pydantic validation error back into the prompt and reasks the model: the same loop with a schema validator in place of the language check.

5.4.5 *Notes.*

- Distinct from its neighbours in *Trusting LLM Output. Defensive Output Parsing* handles output that is malformed rather than wrong, and recovers what it can from the response in hand. A stable, well-understood defect is better fixed in place, with no second call at all. *LLM-Checking-LLM* spends a second LLM call on the check, whereas here the check is classical and nearly free.
- The error-informed retry is the part that generalizes beyond a classical checker: the same move applies whenever a check of any kind rejects a generation, which is why *LLM-Checking-LLM* now names it as one of its failure policies.
- The terminal record is not *LLM Content Validation Tracking*. That pattern tracks the trust level of content that exists; this records that content will never exist for a given input, so the work is not attempted again. It matters most under *Anticipatory Precomputation*, where a scheduled job would otherwise rediscover the same refusal on every run.
- A refusal and a rejected generation are not the same event. The check fires on output the generator produced; a refusal means there is no output to check. The trigger differs, the terminal branch is the same.
- *Maturity.* Four call sites across three Zeeguu features (assessment and summary, article simplification, audio lesson scripts), each with its own policy for what happens when the attempts run out, plus one productized enabler in Instructor. The retry loop was extracted into a shared helper, `generate_in_language`, once a third call site needed it; the helper deliberately does not decide the exhaustion policy, because the call sites disagree about it.

[28] https://python.useinstructor.com/

6 Managing Change Over Time

An LLM integration is not static: prompts and models improve, vendors retire dated snapshots on their own schedule, and the LLM's own role in a feature shifts as the system matures. The patterns in this section manage that change: stamping stored output with the model and prompt that made it so the stale can be found and regenerated, retiring stale artifacts without breaking the past, and treating the LLM as a temporary stand-in for a cheaper component still to be built. The unifying force is a fast-moving, vendor-controlled substrate sitting under long-lived data.

6.1 LLM Output Provenance MANAGING CHANGE PATTERN

Tag every stored LLM artifact with the model and prompt that made it.

6.1.1 Context. LLM-generated artifacts (example sentences, summaries, labels) are written to persistent storage and reused for a long time, while the models and prompts that produce them keep improving. The prompt changes more often than the model, and can matter as much to output quality, sometimes more.

6.1.2 Examples. When the system generates example sentences for a word, it stamps each stored sentence with a `created_by` value naming the model and prompt version that produced it (for example, `claude-opus / examples-v3`). When the example-generation prompt is improved to `v4`, the stale sentences are exactly those still stamped `v3`, so a single query finds them and they are regenerated, without touching the rest of the store.

The stamp is not confined to example sentences. Simplified articles and their per-level summaries, audio lessons, and meaning-frequency labels all carry it, and they carry it by pointing at a shared generator row rather than each repeating a model name in a string column of its own.[29]

That shared entity is what turns regeneration into a join. Asking which artifacts a given prompt version produced is then one lookup, answered across every kind of artifact at once. With the model name denormalized into each table it is four scans over four independently spelled strings, and each new kind of generated artifact adds a fifth.

6.1.3 Problem. When a prompt or model improves, how can exactly the stale artifacts be found and regenerated, without reprocessing the entire store?

6.1.4 Forces.

- **Selective regeneration needs to know how each artifact was made.** Without a record of the model and prompt behind a stored artifact, applying an improved prompt means reprocessing everything. *(pushes toward stamping provenance)*
- **A stamp is only useful if it stays truthful.** Provenance has to be written on every artifact and kept in lockstep with the code: a prompt edited in place without bumping its version leaves the stamp naming the wrong one, silently lying, which is worse than no stamp at all. Keeping it honest is an ongoing discipline, not a one-off write. *(pushes toward stamping precisely what drives regeneration, and versioning it rigorously)*
- **The prompt is the higher-churn axis**: it changes more often than the model and can change the output more, so the stamp must capture the prompt version, not just the model.

6.1.5 Solution. Store the full provenance tuple alongside every LLM-generated artifact: **model version, prompt version, generated output, timestamp**. This turns regeneration into a targeted query, as broad as *"re-run everything produced by prompt v2 with the improved v3"* or as narrow as a single *(model, prompt)* pair, so when a prompt does well on one model and poorly on another, only that combination is redone.

[29] The `AIGenerator` entity (`model_name`, `prompt_version`) and the tables referencing it in the `zeeguu/api` repository.

example_sentence

sentence	created_at	ai_generator_id
Mein Hamster lügt.	2026-03-02	7
Die Katze kündigt.	2026-03-02	7
Das Brot ist müde.	2026-05-10	9

ai_generator

id	model_name	prompt_version
7	opus-4.5	example-gen-v3
9	opus-4.5	example-gen-v4

Fig. 10. Provenance as Zeeguu stores it. Each generated example sentence keeps its own timestamp and points at a shared `ai_generator` row naming the model and prompt version that produced it, rather than repeating them in every row. Finding what a given prompt produced is then a join on one table.

6.1.6 *Consequences.*

- **Selective regeneration becomes a query.** Re-run everything a given prompt or model produced and leave the rest, and the same record doubles as a quality-audit trail.
- **The guarantee lasts only as long as the discipline.** Selective regeneration is trustworthy only while every write stamps accurately; once a stamp drifts from what actually produced the artifact, queries built on it quietly return the wrong rows.
- **One identifier, shared with selection and validation.** The stamped model identifier and the one used to select the model at call time should be the same central constant, kept in one place, and provenance pairs with *LLM Content Validation Tracking*: how an artifact was made, and whether it has been confirmed.

6.1.7 *Known Uses.*

- *Better attested in tooling than in production self-reports.* The capability is productized: MLflow Prompt Registry[30] and LangSmith[31] version prompts and record which prompt/model produced each output, confirming the pattern's core claim that the *prompt* deserves versioning as much as the model, but these are tools, not documented in-app deployments.
- *Adjacent production pipelines.* DoorDash[32] stores versioned LLM-generated profiles and "treat[s] prompts as code"; Etsy[33] stores Pydantic-validated LLM attribute extractions over 100M+ listings. Both store LLM artifacts at scale and version prompts, but neither publicly describes stamping *each artifact* with its prompt version to drive *selective* regeneration: the load-bearing mechanic here. We did not find a first-hand account of the full pattern, so Zeeguu is our instance.

6.1.8 *Notes.*

- The key insight is that the prompt is at least as important to version as the model: a prompt change can completely alter output format, quality, or behaviour even with the same model.
- This is also critical for *Rent, Then Build*: when accumulating LLM-generated labels as training data for a classical replacement, provenance tracking lets one exclude data produced by a prompt version that was later found to be noisy or biased.

[30] https://mlflow.org/docs/latest/genai/prompt-registry/
[31] https://docs.langchain.com/langsmith/prompt-engineering-concepts
[32] https://careersatdoordash.com/blog/doordash-profile-generation-llms-understanding-consumers-merchants-and-items/
[33] https://www.etsy.com/codeascraft/understanding-etsyas-vast-inventory-with-llms

- *Implicit provenance* keeps model names and prompt versions as constants in code and, when the origin of a row is needed, correlates its `created_at` timestamp with git history to find which model/prompt was deployed then. This works only where a single model/prompt is active at a time. Once more than one is live at once (A/B tests, multi-model routing, or the second model of an *Escalate to the LLM* path), the timestamp no longer identifies a unique version and provenance must be stamped explicitly. Explicit tracking also makes analysis faster and keeps the data self-describing.
- *Gateways log calls, not artifacts.* An LLM gateway records each request and response, but selective regeneration needs the provenance to live on the stored artifact and reach the database, where a query can find the stale rows; request logs alone do not drive it.

6.1.9 War Story: The Nine Hundred Lessons. Provenance has to capture the dimension that actually varies. Zeeguu's `audio_lesson_meaning` rows carry a `created_by` field naming the model (`"Claude-Opus-Prompt1"`), but the prompt templates were edited in place without bumping that identifier, so one value spanned two materially different prompt eras. When the ~900 lessons generated under the earlier, ambiguous prompt had to be found, no query could find them. The only route was a content regex on the generated script itself, asking whether it contained the ambiguous phrasing. A provenance field that does not move when the prompt moves is decorative, and selective regeneration degrades into forensics on the output. If prompts evolve by in-place edits, the field naming them must bump on every edit, through a versioned filename or a content hash.

Finding them was half the problem. The rows were also a cache, and they were referenced from lessons learners had already listened to, so they could be neither regenerated in bulk (expensive, and wasted on content never requested again) nor deleted (old daily lessons resolve them by id). Each affected row was given a `deprecated_at` timestamp instead, and the cache lookup was gated to skip deprecated rows: new lessons regenerate under the improved prompt while existing ones keep playing what the learner heard before, and regeneration is paid lazily, one row at a time, on next demand.

That fix contained a second bug. Each lesson's audio file was named after the vocabulary item it taught, which does not change when the lesson is regenerated, so the first regeneration wrote its audio over the recording the deprecated row still needed. Files had to be keyed on the row's own id. Where old artifacts must stay resolvable, every stored by-product has to be keyed by the artifact that produced it, not by the subject it is about.

Both halves trace to one root: the stored artifact did not carry enough of its own history.

6.2 Rent, Then Build MANAGING CHANGE PATTERN

Ship a feature on an LLM now, while building the cheaper component that will replace it.

6.2.1 Context. A feature needs to work in production now, but the efficient, dedicated version of it (a fine-tuned or classical model) needs training data or engineering that does not exist yet. A general-purpose LLM can do the task immediately, if expensively.

6.2.2 Examples. Zeeguu estimates the difficulty level of every article with an LLM, an expensive call it makes constantly. This is also the *bootstrapping* case: those LLM-generated difficulty labels are accumulating as the training set for a cheaper classical classifier that will take the task over once enough have been gathered. The LLM ships the feature and earns its keep today, while quietly producing the data that will one day retire it.

A plainer instance without the bootstrapping twist: article topic classification runs on an LLM now, to be replaced by a dedicated topic-detection model once the taxonomy of available topics settles.

The two tasks are waiting on different things, which is what determines how long each rental lasts. Difficulty assessment is waiting on *volume*, and every call it makes brings the replacement closer, so the stand-in shortens its own tenancy. Topic classification is waiting on a *decision* about the taxonomy, which no amount of LLM traffic advances. A rented capability that generates its own replacement has an end in sight; one that does not can run indefinitely without anything going visibly wrong.

6.2.3 Problem. How can a feature ship and start earning its keep before the cheaper, dedicated version that will eventually run it has been built?

6.2.4 Forces.

- **Shipping now captures the feature and its usage immediately.** A general-purpose LLM can do the task the day it is needed, before any dedicated version exists. *(pushes toward renting)*
- **Running the LLM indefinitely is costly and cedes control.** The feature keeps paying per call, and a core capability stays rented from a vendor who prices and deprecates on their own schedule. *(pushes toward building a replacement)*
- **The replacement cannot be built yet.** A fine-tuned or classical version needs training data or engineering that does not exist at the outset; only running the LLM produces it.

6.2.5 Solution. Use the LLM to perform a task in production while building a more efficient replacement. The arc is rent, then build: the LLM's general capability is *rented* (paid per call, costly but available immediately) to ship the feature now, while a cheaper, dedicated implementation is *built* to eventually own the task. The rented LLM is convincing to users and good for early beta-testing and feedback, but intended to be temporary.

Bootstrapping variant: In the strongest form, the LLM *generates the training data for its own replacement* (as with the difficulty labels above): the rented capability directly produces the labeled data the dedicated successor is trained on, so running the stand-in funds and enables the build.

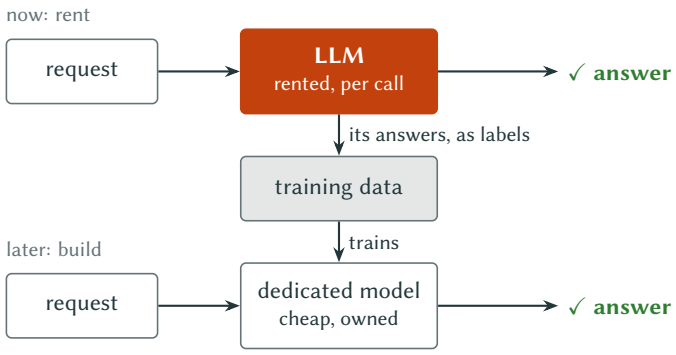


Fig. 11. Now, a rented LLM serves the feature, paid per call. Later, a cheaper dedicated model owned by the system takes the task over. In the bootstrapping variant drawn here, the LLM's answers become the training data for the model that replaces it.

6.2.6 Consequences.

- **The feature is live from day one.** It gathers real usage and feedback immediately, with the LLM standing in for a component that does not exist yet.
- **Running the stand-in funds its successor.** The LLM's inputs and outputs accumulate as the labeled data the dedicated replacement needs, so the temporary solution also produces the training set for the permanent one (the bootstrapping variant).
- **The stand-in is expensive, and "temporary" can stick.** Until the replacement ships, the feature runs at LLM cost and latency, the replacement is a second system to build and validate, and the intended-temporary LLM can quietly become permanent.

6.2.7 Known Uses.

- **OpenAI Model Distillation**[34] (Stored Completions) captures a large model's production input–output pairs and fine-tunes a smaller model as a drop-in replacement: the pattern shipped as a product feature.
- **Self-Instruct / Alpaca [22]** (Wang et al.; Taori et al., 2023) use a strong LLM to generate the instruction data that fine-tunes a small replacement: the bootstrapping variant.
- **Distilling Step-by-Step [9]** (Hsieh et al., ACL Findings 2023) trains task-specific models up to 700× smaller from LLM-generated rationales.

These are distillation and self-instruct methods from the literature; we did not find a first-hand account of a production feature completing the full rent-then-build arc, and Zeeguu's own replacement is still in progress.

6.2.8 Notes.

- This pattern has a lifecycle relationship with *Escalate to the LLM*: a system may start with the LLM as primary (renting it), migrate to a specialized tool as primary, and then keep the LLM as the escalation path, completing a full cycle from LLM-first to LLM-on-demand.

[34] https://openai.com/index/api-model-distillation/

- *Depends on LLM Output Provenance* for the bootstrapping variant: because the replacement trains on the rented LLM's own outputs, those labels must be filterable by the prompt and model version that produced them, so labels from a prompt version later found noisy or biased can be excluded from the training set.
- *Composes with LLM Content Validation Tracking* in the bootstrapping variant: gating the training set to validated outputs (confirmed, not merely generated) lets the dedicated successor learn from checked labels rather than the stand-in's unverified guesses.

7 What Makes These Patterns LLM-Specific?

Some of these patterns echo general distributed systems wisdom: batching (as in *Prompt Amortization*), tiered service, a cheap tier first and a costly one on demand (as in *Escalate to the LLM*), and the redundant dispatch of Zeeguu's parallel translation providers (not catalogued here as a pattern of its own). What makes them distinctly relevant to LLM integration is the combination of forces that arise when an LLM's properties meet the demands of a live system:

- **Cost structure**: per-token pricing with high fixed prompt overhead, unlike flat-rate API calls (drives *Prompt Amortization*, *Escalate to the LLM*, *Anticipatory Precomputation*).
- **Non-determinism**: the same input can yield a different, or malformed, output, so correctness must be enforced around the model (drives *Defensive Output Parsing*, *LLM-Checking-LLM*, *LLM Content Validation Tracking*).
- **Asymmetry between generation and verification**: checking one property is easier than producing the whole output (drives *LLM-Checking-LLM*).
- **General-purpose capability**: the same component can serve as prototype, primary, or fallback (drives *Rent, Then Build* and *Escalate to the LLM*).
- **A rapidly evolving, vendor-controlled substrate**: models and prompts improve and are deprecated on the vendor's schedule, underneath long-lived data (drives *LLM Output Provenance*).
- **Quality-cost-latency tradeoff space**: uniquely wide compared to traditional APIs, and what the efficiency patterns navigate.

8 Related Work

To our knowledge, no peer-reviewed work presents a catalogue of architectural patterns for integrating LLMs as components into existing production systems, grounded in real deployment experience and described using the standard pattern format (context, forces, solution, consequences). The contribution is the catalogue itself, organised by three themes: using the LLM efficiently, trusting its output, and managing change over time. Some patterns apply established mechanisms (batching, precomputation, recall gates, soft-delete, distillation) to the distinctive forces of LLM integration; others, notably *LLM Output Provenance*, *LLM Content Validation Tracking*, and the external-signal trigger of *Escalate to the LLM*, appear to be new. We flag each honestly as one or the other throughout.

Why the gap exists. An absence is worth accounting for. Three things explain this one. The practice is young: production LLM integration became affordable and reliable enough for small teams only around 2023, so the systems this paper describes are roughly three years old, and a pattern needs several years of independent reinvention before anyone can recognise it as one. The knowledge sits with practitioners rather than researchers, and the systems that hold it are commercial, so what is written down appears as engineering blog posts and conference talks rather than papers, and the systems themselves are rarely open to inspection. And the academic attention that does exist has gone elsewhere: the large surveys treat LLMs as tools *for* software engineering, for code generation, repair and testing (Fan et al., 2023 [4]; Zhang et al., 2024 [24]), not as components inside the software being engineered. The one systematic review of software architecture and LLMs found 18 relevant papers in total and called the area open (Schmid et al., 2025 [20]).

This is also why our own corroborating evidence leans on grey literature, and why we think that is defensible rather than a weakness: Garousi et al. (2019) [6] argue for admitting it exactly where the formal literature is silent, practitioners hold the knowledge, and the field moves faster than the publication cycle. All three hold here.

Several practitioner-oriented resources discuss patterns for building LLM-based systems, but they operate at a different level of abstraction and lack the grounding in a specific production system that we aim to provide.

8.1 Practitioner resources

Eugene Yan's **Patterns for Building LLM-based Systems & Products [23]** (2023, blog post) identifies seven patterns: Evals, RAG, Fine-tuning, Caching, Guardrails, Defensive UX, and Collecting User Feedback. These address the question "how do I build an LLM product?" They describe the overall stack for LLM-native applications. Our patterns address a different question: "I have an existing system, and I want to add LLM capabilities as a component: how do I manage cost, quality, latency, and lifecycle?" There is some overlap on caching/pre-computation, but our treatment focuses on prompt amortization and user-need anticipation rather than semantic similarity caching. Yan's work is not peer-reviewed.

ThoughtWorks' **Emerging Patterns in Building GenAI Products [5]** (Fowler et al.) and "Engineering Practices for LLM Applications" focus on operational concerns: testing, evaluation, guardrails, and RAG pipelines. These are primarily about LLMOps rather than the architectural decisions for embedding LLMs as components within an existing application.

Andreessen Horowitz's **Emerging Architectures for LLM Applications [1]** (2023) provides a reference architecture for the LLM infrastructure stack (embedding pipelines, vector databases, orchestration layers, agents). Again, this targets LLM-native products rather than LLM integration into existing systems.

Books. "**LLM Design Patterns**" (Huang, Packt, 2024) and "LLMs in Enterprise" (Menshawy & Fahmy, Packt, 2025) cover model-level patterns (fine-tuning, quantization, inference optimization, RAG) and enterprise deployment concerns. They do not address application-level integration patterns such as the lifecycle management (Rent, Then Build → specialized tool → Escalate to the LLM), prompt amortization, or LLM output provenance that we identify.

8.2 Academic surveys

There is a large body of work on **using LLMs *for* software engineering tasks**: code generation, bug repair, testing, requirements engineering (see surveys by Fan et al., 2023 [4]; Zhang et al., 2024 [24]). However, these focus on LLMs as tools for developers, not on the engineering challenges of integrating LLMs as runtime components within production software.

A recent **systematic literature review on software architecture and LLMs** (Schmid et al., 2025 [20]) found only 18 relevant papers and noted that LLM-based software design remains an open research direction. None of the surveyed academic work addresses the specific architectural patterns for managing cost, quality, latency, and lifecycle when LLMs serve as components in existing applications.

LLM self-verification. For the *LLM-Checking-LLM* pattern, there is relevant work on **LLM self-verification**. Gero et al. (2023) [7] demonstrated that self-verification improves clinical information extraction accuracy, explicitly building on the asymmetry between verification and generation. However, Stechly et al. (2024) [21] showed that self-critique fails for formal reasoning tasks, finding significant performance collapse with self-verification but gains with external verification. Our pattern differs from both in that it uses separate, differently-prompted LLM calls for generation and verification of different properties (e.g., text simplification followed by grammatical correction), rather than asking an LLM to verify its own reasoning.

9 Limitations and Future Work

The patterns in this catalogue have been extracted from a single production system, Zeeguu, in the language-learning domain. We have argued throughout that these forces arise from the LLM's own properties (per-token cost, multi-second latency, non-determinism, imprecision, a rapidly shifting provider landscape, and general-purpose capability) meeting the demands of a live application, neither of which is specific to language learning, and that the patterns should therefore generalise to other interactive applications that surface LLM-generated text to end users. Whether this is borne out in practice is an empirical question we cannot fully answer from a single case study. The most direct way to validate or refute the catalogue is to gather complementary examples (and counter-examples) from practitioners working in other domains. A PLoP writers' workshop is one venue for exactly this kind of community refinement; mining open-source repositories for additional instances of these patterns (and patterns we missed) is a natural follow-up.

A further threat is to the premise rather than to the patterns. We expect LLMs to sit behind existing features more and more often, but assistants running on dedicated hardware could instead become where people go for everything, reducing applications to interfaces those assistants call. On that reading this paper describes a transitional form. We do not believe this will happen very soon but if it does, indeed, the relevance of these patterns might decrease.

A second limitation is that the catalogue reports patterns we have found useful but does not yet quantify their impact systematically. Cost and latency improvements are reported informally for some patterns; a more rigorous deployment-level evaluation (measuring, for example, how much of Zeeguu's LLM bill is attributable to which patterns, or how much user-perceived latency racing providers removes) is future work.

A further direction is to treat the catalogue as the seed of a *pattern language* rather than a flat list. The patterns are not independent: they compose (pre-computation feeds *Prompt Amortization*; *LLM Output Provenance* records how an artifact was made and *LLM Content Validation Tracking* whether it has since been confirmed) and they evolve over a system's lifetime (an LLM may begin as a *Rent, Then Build* stand-in, hand off to a specialized tool, and remain as the *Escalate to the LLM* fallback). Documenting these interactions, sequences, and tensions, so a designer can navigate from one pattern to the next, is a contribution beyond the individual patterns.

These choices are not made once. A system's pattern mix is actively revised as it scales and the provider landscape shifts: Zeeguu began by making single LLM calls and only recently added parallel, raced ones, and may later drop that approach or fold it together with per-user budgeting as usage grows. A longitudinal account of how and why a system's chosen patterns change over time would deepen the lifecycle dimension that this single-snapshot catalogue can only sketch.

Finally, we expect the catalogue to be incomplete[35]: further patterns will likely surface only through engagement with practitioners working at different scales, in different domains, and with different LLM providers.

10 Conclusion

This paper has presented a catalogue of architectural patterns for integrating LLMs as components into existing user-facing applications: a setting that has received much less attention than the design of LLM-native products, despite arguably affecting a larger share of working software engineers. The patterns address recurring concerns around cost, latency, lifecycle, data management, and quality assurance, and were drawn from real production experience on the Zeeguu platform. They are intended as a starting point for community refinement rather than a closed taxonomy:

[35] The patterns in this paper are a selection from a larger, evolving catalogue maintained online at https://mircealungu.github.io/llm-integration-patterns/

practitioners working on LLM integration in other domains are warmly invited to validate, refute, extend, or replace individual patterns, and to contribute new ones the catalogue does not yet contain.

11 Acknowledgments

We thank Viktor Matovič, our shepherd at PLoP, for feedback that strengthened the paper throughout, and above all for his insistence that patterns be observed rather than asserted, which improved our methodology and ultimately changed the catalogue itself. We thank Souhaila Serbout, Oscar Nierstrasz, and Diomidis Spinellis for feedback and discussions on earlier versions of this work. We are also grateful to the participants of the EuroPLoP focus group, whose discussion sharpened several patterns, from their naming to their relationship with both older, non-LLM patterns and the emerging LLM-gateway infrastructure.

References

[1] Andreessen Horowitz. 2023. Emerging Architectures for LLM Applications. https://a16z.com/emerging-architectures-for-llm-applications/

[2] Lingjiao Chen, Matei Zaharia, and James Zou. 2023. FrugalGPT: How to Use Large Language Models While Reducing Cost and Improving Performance. *Transactions on Machine Learning Research (TMLR)* (2023). https://arxiv.org/abs/2305.05176

[3] Zhoujun Cheng, Jungo Kasai, and Tao Yu. 2023. Batch Prompting: Efficient Inference with Large Language Model APIs. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: Industry Track (EMNLP)*. https://arxiv.org/abs/2301.08721

[4] Angela Fan, Beliz Gokkaya, Mark Harman, Mitya Lyubarskiy, Shubho Sengupta, Shin Yoo, and Jie M. Zhang. 2023. Large Language Models for Software Engineering: Survey and Open Problems. In *IEEE/ACM International Conference on Software Engineering: Future of Software Engineering (ICSE-FoSE)*. https://arxiv.org/abs/2310.03533

[5] Martin Fowler et al. 2024. Emerging Patterns in Building GenAI Products. https://martinfowler.com/articles/gen-ai-patterns/

[6] Vahid Garousi, Michael Felderer, and Mika V. Mäntylä. 2019. Guidelines for including grey literature and conducting multivocal literature reviews in software engineering. *Information and Software Technology* 106 (2019), 101–121. doi:10.1016/j.infsof.2018.09.006

[7] Zelalem Gero, Chandan Singh, Hao Cheng, Tristan Naumann, Michel Galley, Jianfeng Gao, and Hoifung Poon. 2023. Self-Verification Improves Few-Shot Clinical Information Extraction. In *ICML 3rd Workshop on Interpretable Machine Learning in Healthcare (IMLH)*. https://arxiv.org/abs/2306.00024

[8] Alex Gilmore. 2007. Authentic materials and authenticity in foreign language learning. *Language Teaching* 40, 2 (2007), 97–118. https://doi.org/10.1017/S0261444807004144

[9] Cheng-Yu Hsieh, Chun-Liang Li, Chih-Kuan Yeh, Hootan Nakhost, Yasuhisa Fujii, Alexander Ratner, Ranjay Krishna, Chen-Yu Lee, and Tomas Pfister. 2023. Distilling Step-by-Step! Outperforming Larger Language Models with Less Training Data and Smaller Model Sizes. In *Findings of the Association for Computational Linguistics (ACL)*. https://arxiv.org/abs/2305.02301

[10] Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. 2023. Large Language Models Cannot Self-Correct Reasoning Yet. https://arxiv.org/abs/2310.01798

[11] Omar Khattab and Matei Zaharia. 2020. ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*. https://arxiv.org/abs/2004.12832

[12] Stephen D. Krashen. 1982. *Principles and Practice in Second Language Acquisition*. Pergamon Press. https://sdkrashen.com/content/books/principles_and_practice.pdf

[13] Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. 2023. G-Eval: NLG Evaluation using GPT-4 with Better Human Alignment. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. https://arxiv.org/abs/2303.16634

[14] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, et al. 2023. Self-Refine: Iterative Refinement with Self-Feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*. https://arxiv.org/abs/2303.17651

[15] Leon Moonen. 2001. Generating Robust Parsers Using Island Grammars. In *Proceedings of the Eighth Working Conference on Reverse Engineering (WCRE)*. 13–22. https://doi.org/10.1109/WCRE.2001.957806

[16] Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M. Waleed Kadous, and Ion Stoica. 2024. RouteLLM: Learning to Route LLMs with Preference Data. *arXiv preprint arXiv:2406.18665* (2024). https://arxiv.org/abs/2406.18665

[17] Peng Qi, Yuhao Zhang, Yuhui Zhang, Jason Bolton, and Christopher D. Manning. 2020. Stanza: A Python Natural Language Processing Toolkit for Many Human Languages. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL): System Demonstrations*. https://arxiv.org/abs/2003.07082

[18] Alexander Ratner, Stephen H. Bach, Henry Ehrenberg, Jason Fries, Sen Wu, and Christopher Ré. 2017. Snorkel: Rapid Training Data Creation with Weak Supervision. *Proceedings of the VLDB Endowment* 11, 3 (2017). https://arxiv.org/abs/1711.10160

[19] William Saunders, Catherine Yeh, Jeff Wu, Steven Bills, Long Ouyang, Jonathan Ward, and Jan Leike. 2022. Self-critiquing models for assisting human evaluators. https://arxiv.org/abs/2206.05802

[20] Larissa Schmid et al. 2025. Software Architecture Meets LLMs: A Systematic Literature Review. (2025). https://arxiv.org/abs/2505.16697

[21] Kaya Stechly, Karthik Valmeekam, and Subbarao Kambhampati. 2024. On the Self-Verification Limitations of Large Language Models on Reasoning and Planning Tasks. https://arxiv.org/abs/2402.08115

[22] Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. 2023. Stanford Alpaca: An Instruction-following LLaMA Model. https://github.com/tatsu-lab/stanford_alpaca. https://github.com/tatsu-lab/stanford_alpaca

[23] Eugene Yan. 2023. Patterns for Building LLM-based Systems and Products. https://eugeneyan.com/writing/llm-patterns/

[24] Quanjun Zhang, Chunrong Fang, Yang Xie, Yuxiang Zhang, Yang Yang, Weisong Sun, Shengcheng Yu, and Zhenyu Chen. 2024. A Survey on Large Language Models for Software Engineering. (2024). https://arxiv.org/abs/2312.15223

[25] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems (NeurIPS)*. https://arxiv.org/abs/2306.05685